Total No. of Questions : 4]

SEAT No. :

PE139

[Total No. of Pages : 2

[6579]-472

T.E. (Cyber Security) (Honors) (Insem)

INFORMATION AND CYBER SECURITY

(2019 Pattern) (Semester-I) (310401)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagram must be drawn whenever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***
5) ***Use of Scientific Calculator is permitted.***

Q1) a) Enumerate and provide explanations for the fundamental components that form the foundation of computer security. **[5]**

b) Distinguish between policy and mechanism with respect to computer security. **[5]**

c) How do we determine whether a policy correctly describes the required level and type of security for the site? **[5]**

OR

Q2) a) Define the concept of a threat and explain the four primary categories that involve a wide range of threats. **[5]**

b) Given a security policy's specification of "secure" and "nonsecure" actions, explain different security mechanism. **[5]**

c) What operational challenges does Information Security face? **[5]**

Q3) a) Explain Data encryption standard with diagram. **[5]**

b) Provide a differentiation between block and chain ciphers, illustrating each with an example and discussing the techniques employed. **[5]**

c) Define "modulus" and elaborate on the concept of congruent modulo along with an exploration of its properties. **[5]**

OR

Q4) a) Explain step by step Advanced encryption standard. **[5]**

b) Enumerate various characteristics of modulus arithmetic. **[5]**

c) Describe Euclid's method for finding the Greatest Common Divisor (GCD) and explain the extended Euclidean algorithm. **[5]**

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